

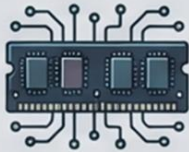
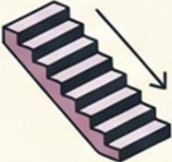
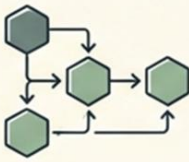


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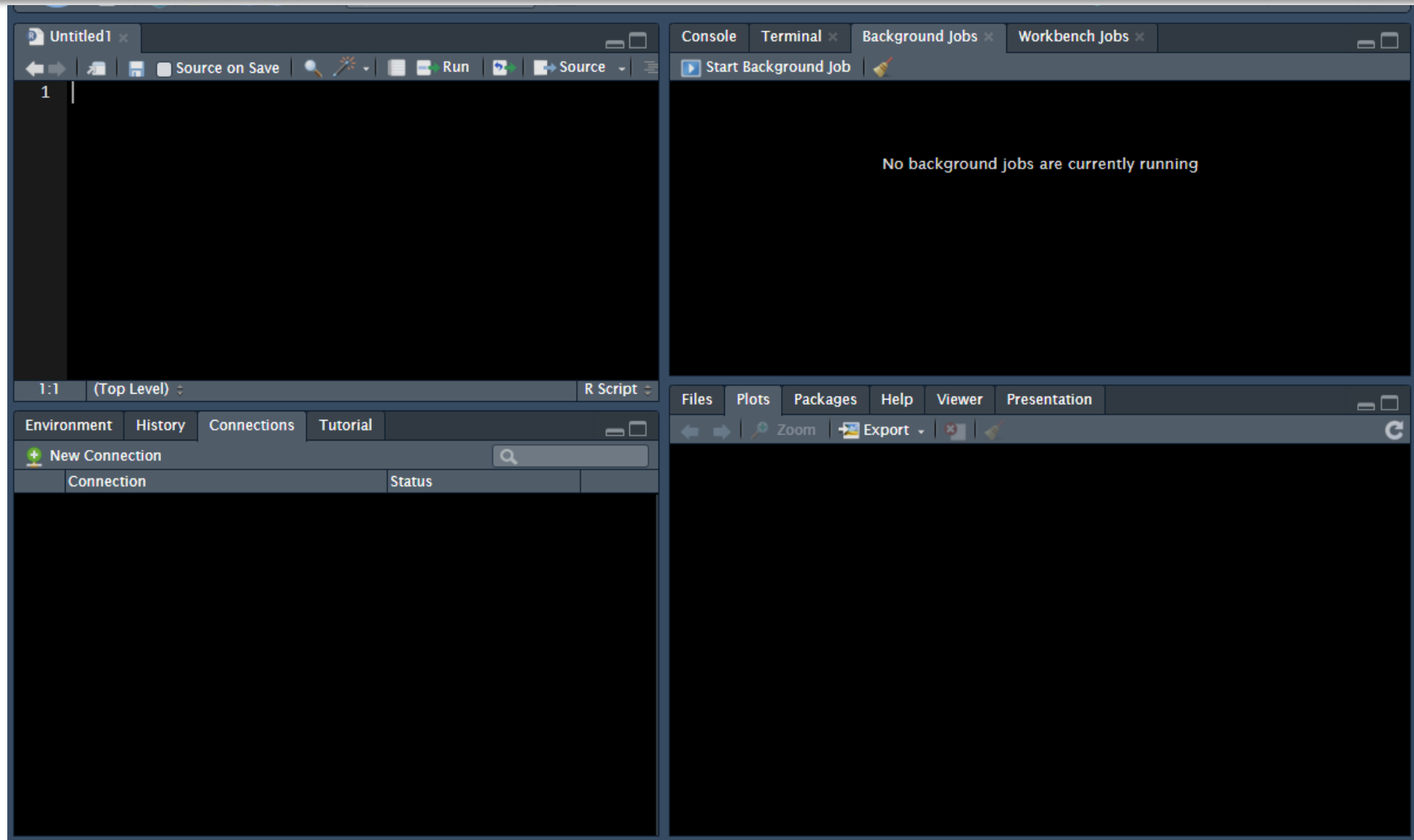
**Jason Geslois, MPH, MAS, EMBA
Senior Epidemiologist**

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Translating the Philosophy: SAS vs. R

Dimension	Legacy (SAS)	Modern (R)
Data Storage	 Disk-based tables. Persistent storage.	 In-memory Data Frames. Active RAM.
Execution	 Sequential PROCs & DATA steps.	 Interactive, object-oriented functions.
Ecosystem	 Proprietary, built-in modules.	 Open-source, community Packages.
Workspace	 Libraries and Libnames.	 Self-contained RStudio Projects.

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The Command Center

 →  Translation Key

Source / Files

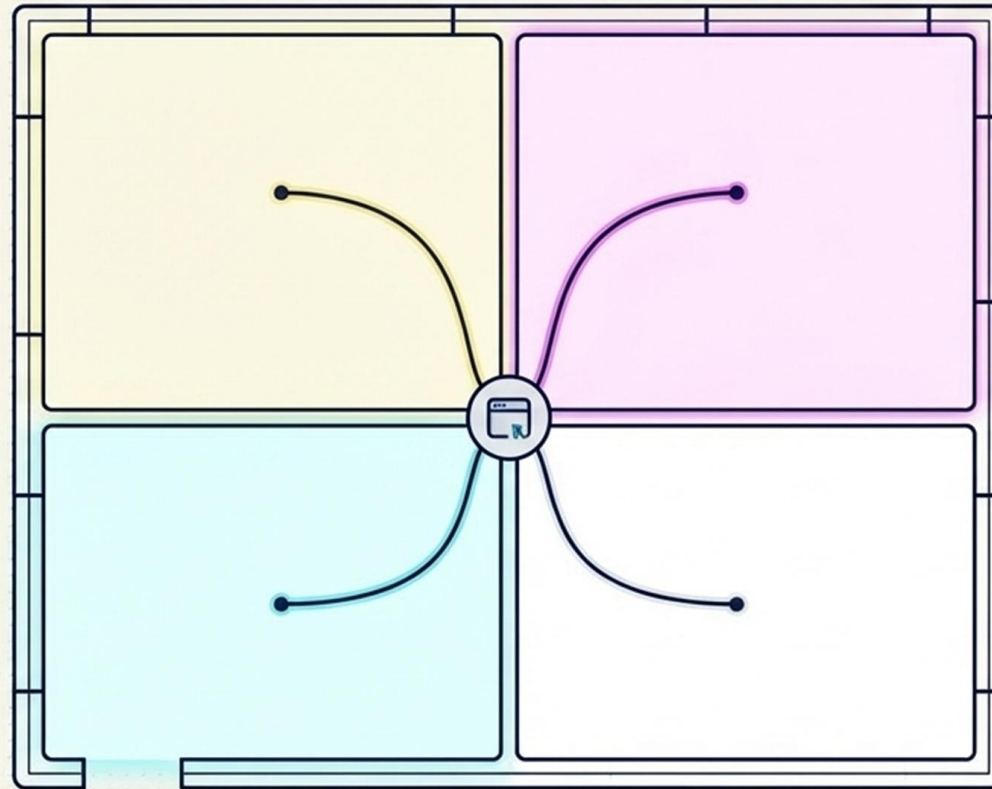
The Drafting Board.

Where scripts, markdown, and quarto files are written.

Console

The Engine. Where code executes instantly.

⚠ Unlike SAS, you don't need a Run command for every line; the console is live.



Environment

Active Memory.

Displays loaded data and saved objects.

Output

The Display. Houses the Plot pane, Help pane, and local file directory.

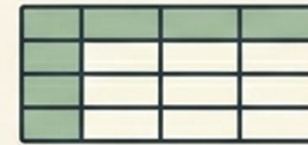
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The Drafting Table (Source)



Where you write reproducible code.

The Workbench (Environment)



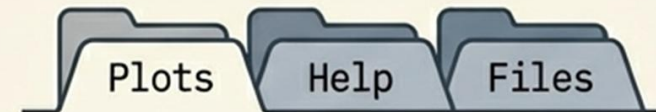
View active objects, loaded Data Frames, and manage memory.

The Engine Room (Console)



For immediate execution and testing code on the fly.

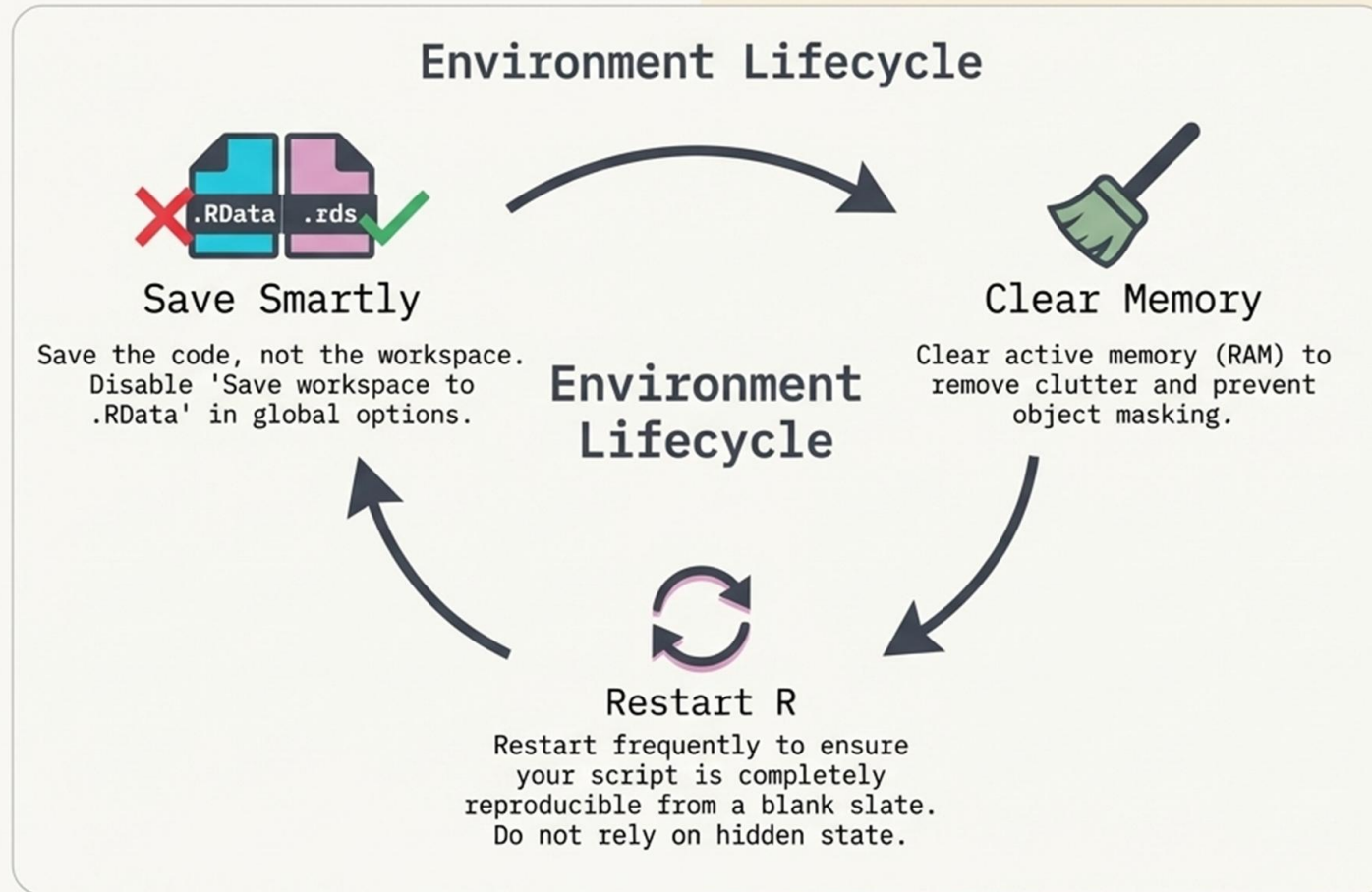
The Display (Output)



Review visualizations, read package documentation, and browse your directory.

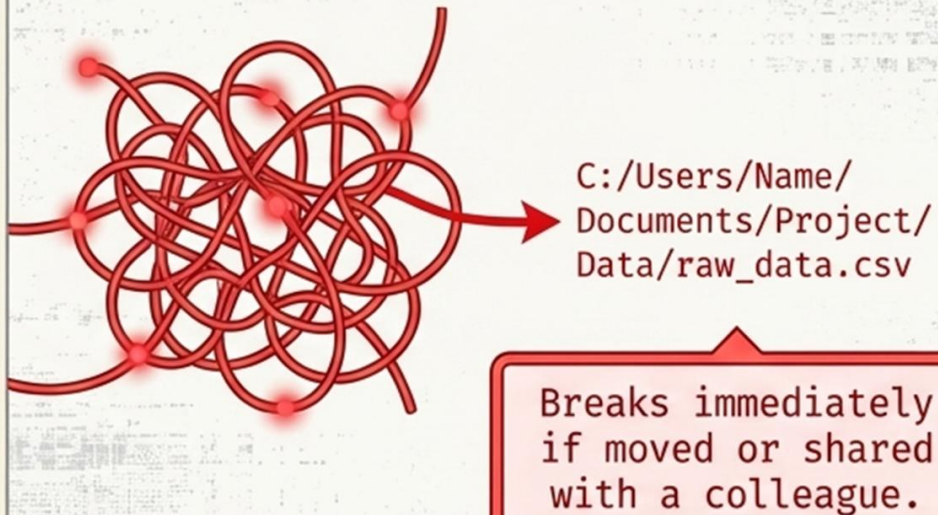
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Session Management and Active Memory

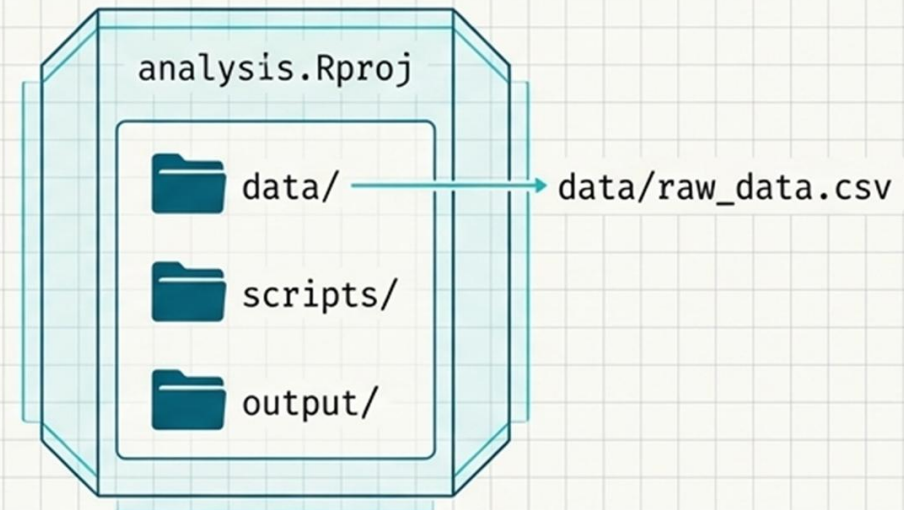


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The Old Way



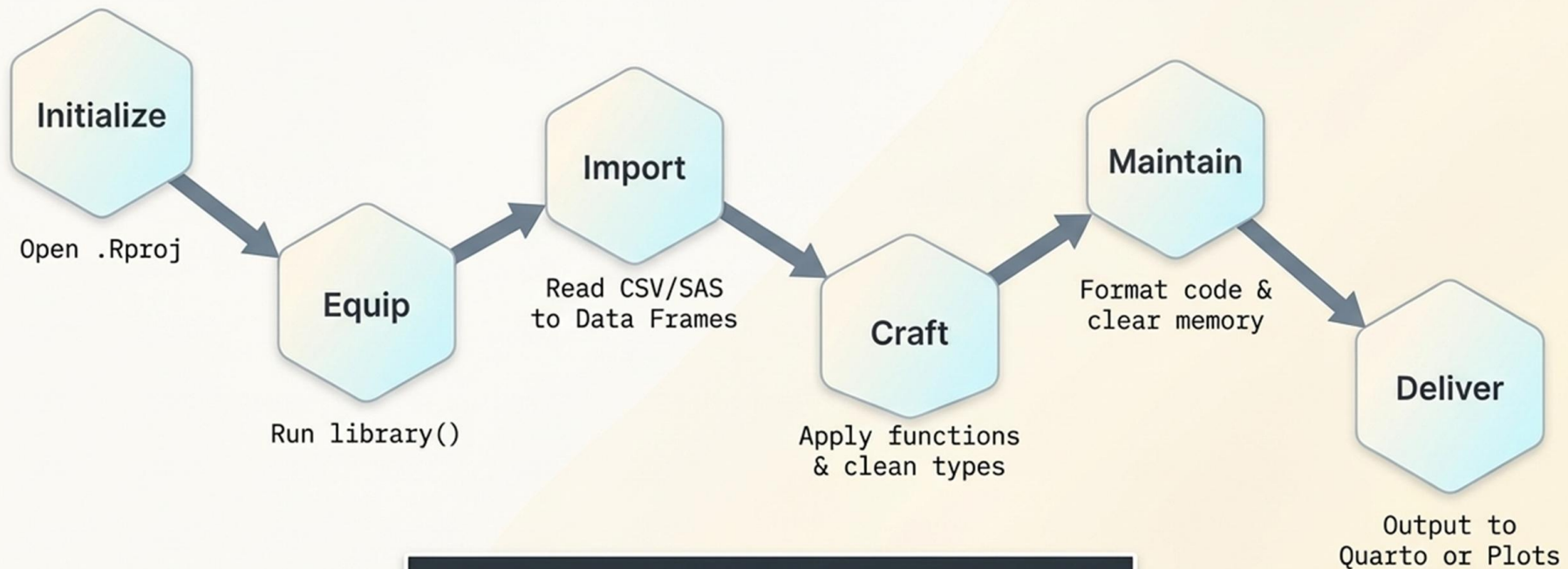
The R Project Way



A Project is a self-contained universe. By using relative paths, the code just works—no matter whose computer opens it.

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Synthesis: The R Workflow Blueprint



You now possess the blueprint. The transition from legacy scripts to reproducible environments begins here.

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Document Decision Matrix

Scripts (`.R`)

Use Case: Backend data processing, simple functions.

Analogy: The rough draft / engine room.



Markdown (`.Rmd`)

Use Case: Interleaving text, code, and plot panes.

Analogy: The standard analyst notebook.



Quarto (`.qmd`)

Use Case: Modern, multi-format publishing (HTML, PDF, slides).

Analogy: The final polished report.



The Modern Standard

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```
{r setup, include=FALSE}  
knitr::opts_chunk$set(echo = TRUE)
```



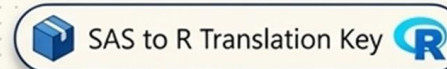
Loading Libraries & Data Directories



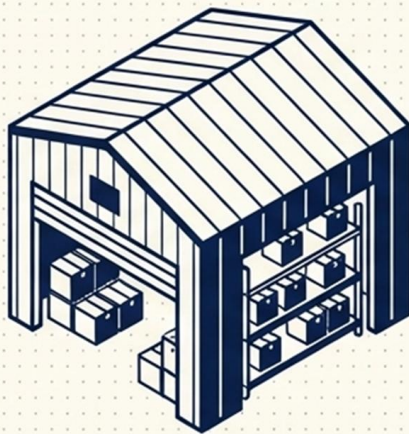
Here is where you first load the packages.

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The Package Supply Chain

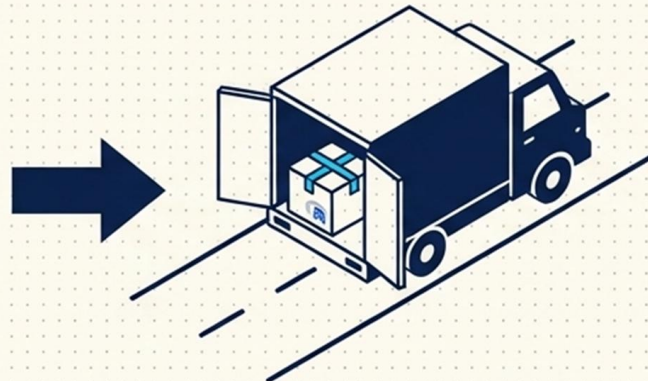


CRAN (The Warehouse)



The global repository of open-source packages.

`install.packages()`



⚠ The Delivery: Do this only once per machine to bring the tool to your hard drive.

`library()`



i The Workbench: Take the tool out of your local library and put it on your active desk. Do this every time you start a new session.

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Rainbow Code

```
(function(data){
  return data.filter(d =>
    (d.value > 10 &&
      (d.type === 'A' ||
        (d.type === 'B' && d.status === 'active'))));
})
```

Rainbow brackets prevent missing parenthesis errors and visually untangle complex nesting.

Find / Replace All

```
// unuse script console
for (variable, session$old_variable) = new_clean_var {
  return return, new_clean_var => old_variable;
}

// 
if (new_variable = session$old_variable) {
  replace(new_clean_var variable, new_clean_var => new_clean_var);
}
```

Find: old_variable → Replace: new_clean_var

Instantly transform variable names across a whole script without hunting.

Status Bar & Global Options

Word Count: 1,420

≡ Hidden Characters

• Markdown/Quarto

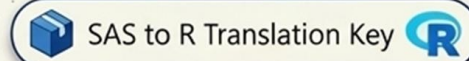
Word Count: 1,420

Line: 87, Col: 15

Enable word counts and hidden character views to optimize writing in Markdown and Quarto.

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Code Craftsmanship



The Messy Block

```
data mess; set raw;
  if x=1 then do;
    y=x*2;
    z=y+3;
    if z>10 then w=z-2; else w=z+2;
  end;
  else do;
    y=x*3;
    z=y+2; if z<5 then w=z/2;
  else w=z*2; end; run;
proc print data=mess; run;
```

The Clean Block

```
# Data Transformation for Final Output

messy_data <-
  raw_data |>
  mutate(
    y = case_when(
      x == 1 ~ x * 2,
      TRUE ~ x * 3
    ),
    z = case_when(
      x == 1 ~ y + 3,
      TRUE ~ y + 2
    ),
    w = case_when(
      z > 10 ~ z - 2,
      z < 5 ~ z / 2,
      TRUE ~ z + 2
    )
  )
```

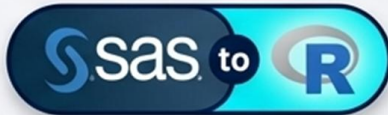
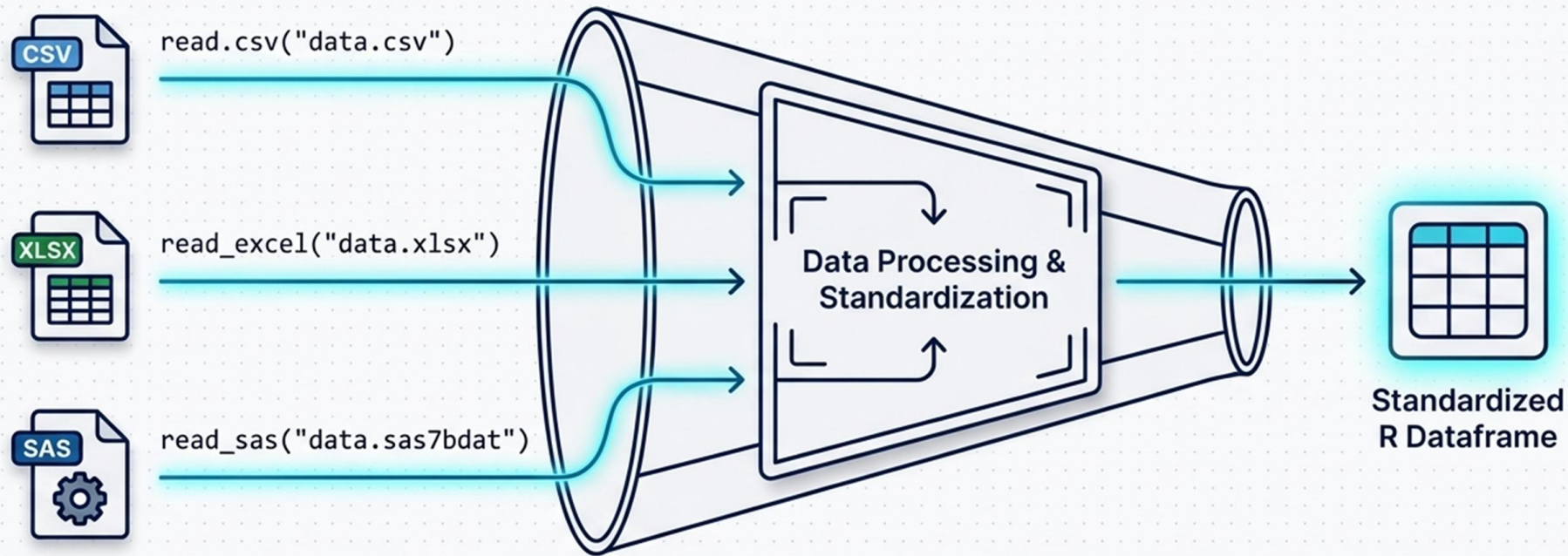
Commenting: Use # for inline context and to divide code into functional sections.

Reflow: Wrapping long text lines for readability.

Re-indent: Automatically aligning nested logic.

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Data Ingestion Pipeline



**SAS to R
Translation Key**

In SAS, you write LIBNAME statements.
In R, you assign reading functions directly to an object.

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Anatomy of R Syntax



SAS

```
PROC MEANS DATA=mydata; VAR myvar; RUN;
```

R

```
summary_data <- calculate_mean(data, na.rm = TRUE)
```

The Environment Object

This pushes
the result into
memory.

The Assignment
Operator

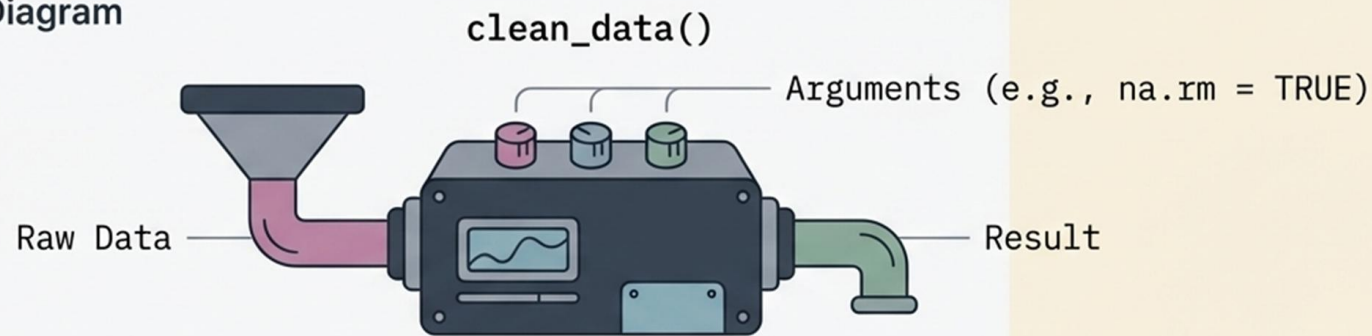
The Simple Function

The Arguments
(controlling function behavior)

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Crafting with Functions

The Blueprint Diagram



Code Comparison

Run 1

```
mean(sales_data, na.rm = FALSE)
```

NA

Run 2

```
mean(sales_data, na.rm = TRUE)
```

42.5

Highlight and run different lines of code sequentially in the Console to instantly compare outputs and tweak function arguments.

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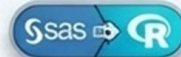
Quick Reference / Cheat Sheet



Top Keyboard Shortcuts

Assignment (<-)	Alt	-
Run Code	Ctrl/Cmd	Enter
Find/Replace All	Ctrl/Cmd	Shift J
Restart Session	Ctrl/Cmd	Shift F10

SAS to R Rosetta Stone Summary



Workspace	→	R Project
LIBNAME	→	read_*() functions
PROC	→	Modular Functions
.sas	→	.R, .Rmd, .qmd

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Keyboard Shortcut Quick Reference

Tabs

- Ctrl+Shift+., Switch to Tab...
- Ctrl+F12 Next Tab
- Ctrl+F11 Previous Tab
- Ctrl+Shift+F11 First Tab
- Ctrl+Shift+F12 Last Tab

Panes

- Ctrl+1 Move Focus to Source
- Ctrl+2 Move Focus to Console
- Ctrl+3 Move Focus to Help
- Ctrl+4 Show History
- Ctrl+5 Show Files
- Ctrl+6 Show Plots
- Ctrl+7 Show Packages
- Ctrl+8 Show Environment
- Ctrl+9 Show Viewer
- Ctrl+F1 Show Vcs
- Ctrl+F2 Show Build
- Ctrl+F5 Show Connections
- Ctrl+F8 Show Presentation
- Ctrl+F6 Show Find in Files
- Ctrl+F7 Add Source Column
- Ctrl+Shift+0 Show All Panes

Add Shift to zoom (maximize) pane.

Files

- Ctrl+S Save
- Ctrl+Alt+S Save All
- Ctrl+Alt+Shift+N New R Script
- Ctrl+O Open File...
- Ctrl+W Close
- Ctrl+Alt+Shift+W Close All Except Current
- Ctrl+Shift+W Close All
- Ctrl+Shift+F Find in Files...

Source Navigation

- Ctrl+F9 Back
- Ctrl+F10 Forward
- Ctrl+Alt+U Find Usages
- Ctrl+F3 Use Selection for Find
- Ctrl+F Find...
- F3 Find Next
- Shift+F3 Find Previous
- Ctrl+Shift+J Replace and Find
- Ctrl+. Go To File/Function...
- Alt+Shift+G Go to Line...
- Alt+Shift+J Jump To...
- Ctrl+P Jump To Matching
- Ctrl+Alt+Shift+E Expand To Matching
- Ctrl+Shift+O Show Document Outline
- Ctrl+Alt+Up Add Cursor Above Current Cursor
- Ctrl+Alt+Down Add Cursor Below Current Cursor
- Ctrl+Shift+Up Expand Selection
- Ctrl+Shift+Down Shrink Selection
- Ctrl+PgDn Go to Next Section
- Ctrl+PgUp Go to Previous Section
- Ctrl+Alt+A Split Into Lines
- Ctrl+Alt+Shift+A Edit Lines from Start
- Ctrl+Shift+F4 Use Visual Editor
- Ctrl+Alt+Shift+Up Move active cursor up
- Ctrl+Alt+Shift+Down Move active cursor down

Execute

- Ctrl+Shift+S Source Active File
- Ctrl+Shift+Enter Source with Echo
- Ctrl+Shift+Enter Preview JS
- Ctrl+Shift+Enter Preview SQL
- Ctrl+Alt+G Source a File...
- Ctrl+Alt+P Re-Run Previous
- Ctrl+Enter Run Selected Line(s)
- Alt+Enter Run Line(s) without moving cursor
- Ctrl+Alt+R Run All

Source Editor

- Ctrl+Alt+I Insert Chunk
- Ctrl+Shift+R Insert Section...
- Ctrl+Alt+X Extract Function
- Ctrl+Alt+V Extract Variable
- Ctrl+Shift+C Comment/Uncomment Lines
- Ctrl+I Reindent Lines
- Ctrl+Shift+/ Reflow Comment
- Ctrl+Shift+A Reformat Selection
- Ctrl+Alt+Shift+D Show Diagnostics (Project)
- Alt+L Collapse Fold
- Alt+Shift+L Expand Fold
- Alt+O Collapse All Folds
- Alt+Shift+O Expand All Folds
- Alt+Up Move Lines Up
- Alt+Down Move Lines Down
- Ctrl+D Delete Line
- Ctrl+U Yank Line Up to Cursor
- Ctrl+K Yank Line After Cursor
- Ctrl+Y Insert Yanked Text
- Alt+- Insert Assignment Operator
- Ctrl+Shift+M Insert Pipe Operator
- Ctrl+Alt+Shift+M Rename in Scope
- Ctrl+Alt+Shift+R Insert Roxygen Skeleton
- Ctrl+Backslash Request Copilot Code Suggestions

Debug

- Shift+F9 Toggle Breakpoint
- F10 Execute Next Line
- Shift+F4 Step Into Function
- Shift+F7 Finish Function/Loop
- Shift+F5 Continue
- Shift+F8 Stop Debugging

Accessibility

- Alt+Shift+/, Screen Reader Support

Source Control

- Ctrl+Alt+D Diff Files
- Ctrl+Alt+M Commit...

Build

No background jobs are currently running

- Ctrl+Shift+K Compile PDF
- Ctrl+Shift+K Preview
- Ctrl+Shift+K Knit Document
- Ctrl+Shift+K Render Document
- Ctrl+Shift+K Run Document
- Ctrl+Shift+B Install Package
- Ctrl+Shift+L Load All
- Ctrl+Shift+E Check Package
- Ctrl+Alt+F7 Test Package
- Ctrl+Shift+D Document

Console

- Ctrl+L Clear Console
- Ctrl+Up Popup Command History

Terminal

- Alt+Shift+R New Terminal
- Alt+Shift+M Move Focus to Terminal
- Alt+Shift+F11 Previous Terminal
- Alt+Shift+F12 Next Terminal

Other

- F1 Show Function Help
- F2 Go To Function / File
- Ctrl+Space Complete Code
- Ctrl+Shift+F10 Restart R
- Ctrl+Alt+F11 Previous Plot
- Ctrl+Alt+F12 Next Plot
- Ctrl+Shift+` Log focused element
- Ctrl+Shift+U Choose Directory

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Now for a Live Demo

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Thank You/Questions

